

Semi-prime factorization – Analysis and improvements on Fermat's Factorization method

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Foreword

This is a work in progress. I need to add a lot more information. However, in this iteration, I hope to compress everything down to as few pages as possible.

No AI was used for this research project. Most of the foundation I uncovered back in 2024 already, but as I had only been studying math for a year, it took me a few additional years to reach the maturity required to make all the right connections. I can however backtrack and explain how I uncovered everything that is described here, from the ground up, with pen and paper, all the way back in 2023 when I started this project.

To do: Add references and closing thoughts and clean everything up.

Chapter I – Fermat's Factorization method

In its most basic form the procedure for Fermat's Factorization method is as follows (where N is the semi-prime being factored).

1. Calculate $\sqrt{N}$ in $\mathbb{Z}$
2. Starting from $x = \sqrt{N}$, calculate $f(x) = x^2 - N$ in a loop, incrementing x .
3. If $f(x)$ evaluates to a square, y^2 , calculate the greatest common divisor (GCD) on the difference of x and y .

Taking $N = 4387$ as an example:

Step 1 (Calculate the +/- square root of N):

$$\sqrt{4387} = 66 \text{ in } \mathbb{Z}$$

Step 2 (Starting from the square root calculate $f(x) = x^2 - N$:

$66^2 - 4387 = -31$ (not a square)
 $67^2 - 4387 = 102$ (not a square)
 $68^2 - 4387 = 237$ (not a square)
 $69^2 - 4387 = 374$ (not a square)
 $70^2 - 4387 = 513$ (not a square)
 $71^2 - 4387 = 654$ (not a square)
 $72^2 - 4387 = 797$ (not a square)
 $73^2 - 4387 = 942$ (not a square)
 $74^2 - 4387 = 1089 = 33^2$ (square)

Step 3 (If the result is also a square, calculate the GCD on the difference.):

$$74^2 = 33^2 \text{ mod } 4387$$

$$\begin{aligned} \text{GCD}(74 + 33, 4387) &= 107 \\ \text{GCD}(74 - 33, 4387) &= 41 \end{aligned}$$

Both the Quadratic Sieve and Number Field Sieve algorithms (to do: insert references) are expansions on this method.

Chapter II – Analysis

a. Binomial expansion

As seen in chapter 1 the goal is the find the following numbers that we can use to take the GCD with:

$$\begin{aligned} \text{GCD}(x + y, N) &= q \\ \text{GCD}(x - y, N) &= p \end{aligned}$$

x and y can thus be rewritten as the differences of the factors of N , $p + q$ or $p - q$.
In addition, many more such solutions exist created by $(ap + bq)/2$ and $(ap - bq)/2$ where a and b are multipliers.

If $a = 13$, $b = 5$ and $N = 4387$ (and thus factors $p = 41$ and $q = 107$)

We get:

$$\begin{aligned} (13 \times 41 + 5 \times 107)/2 &= 534 \\ (13 \times 41 - 5 \times 107)/2 &= -1 \end{aligned}$$

And we can see this creates the following expression when we square 534 modulo N (4387 here):

$$534^2 - 4387 \times 13 \times 5 = 1^2$$

This behavior can be explained using binomial expansion.

I will shorten binomial expansion as $BE(x - y)^e$.
For example $BE(x - 74)^2 = x^2 - 148x + 5476$.

An interesting property of performing these binomial expansions is that we in essence simply increase the exponent on the binomial value.

For example:

$$BE(33 - 74)^1 = 33 - 74 = -41$$

$$\begin{aligned}
BE(33 - 74)^2 &= 33^2 - 148 \times 33 + 5476 = -41^2 \\
BE(33 - 74)^3 &= 33^3 - 222 \times 33^2 + 16428 \times 33 - 405224 = -41^3 \\
BE(33 - 74)^4 &= 33^4 - 296 \times 33^3 + 32856 \times 33^2 - 1620896 \times 33 + 29986576 = -41^4
\end{aligned}$$

Conversely we also see the following occurring if we substitute in the factor as root instead (or swapping a binomial term with the binomial value):

$$\begin{aligned}
BE(41 - 74)^1 &= 41 - 74 = -33 \\
BE(41 - 74)^2 &= 41^2 - 148 \times 41 + 5476 = -33^2 \\
BE(41 - 74)^3 &= 41^3 - 222 \times 41^2 + 16428 \times 41 - 405224 = -33^3 \\
BE(41 - 74)^4 &= 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 29986576 = -33^4
\end{aligned}$$

In addition, if we have even exponents we can substitute the constant by a multiple of N and generate a polynomial p to solve for 0.

$$\begin{aligned}
BE(41 - 74)^2 &= 41^2 - 148 \times 41 + 5476 \rightarrow p = 41^2 - 148 \times 41 + 4387 = 0 \\
BE(41 - 74)^4 &= 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 29986576 \rightarrow p = 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 4387 \times 6565 = 0
\end{aligned}$$

This behavior is rather crucial.

b. The discriminant

Of-course we cannot blindly perform binomial expansion on random numbers. We need some type of measure indicating us how close or far away we are from a solution.

Take for example the following binomial expansion $BE(x - 74)^2 = x^2 - 148x + 5476$ and on this quadratic polynomial p , we substitute the constant with N (for example $N = 4387$):

$$p = x^2 - 148x + 4387$$

If we now calculate the discriminant of p , $DISC(p) = 148^2 - 4 \times 4387 = 66^2$, we see that the result is also square. And we now have the square relation $148^2 = 66^2 \pmod{4387}$, whose $GCD(148 - 66, 4387) = 41$.

An observation:

From this we can make the following generalization that we can derive the factorization of a semi-prime N , if we find a quadratic polynomial with a constant set to N , or a multiple thereof, and whose discriminant yields a square. And from this we can also assume that the discriminant would yield a quadratic residue modulo any prime.

We also observe that the discriminant value is double the square of the binomial term. And can thus also be generated with $BE(x - 74)^2 = x^2 - 148x + 5476$ by substituting the root with a factor of N , as seen earlier. Since $BE(41 - 74)^2 = 41^2 - 148 \times 41 + 5476 = -33^2$. This is critical as calculating discriminant is not trivial as the polynomial degree goes up, but this gives us an alternative way to do it.

c. Using binomial expansion to create an oracle for the correct solution

The below examples assume $N = 4387$.

Previously we had the following binomial b :

$$b = 33 - 74 = -41$$

Next performing binomial expansion to degree 2, if we then substitute the constant by N to create polynomial p , and swap a binomial term with the binomial value (such that $b = 41 - 74$ instead), if p evaluates to 0, we have a valid solution.

$$BE(41 - 74)^2 = 41^2 - 148 \times 41 + 5476 = 33^2 \rightarrow p = 41^2 - 148 \times 41 + 4387 = 0$$

For binomial term $y = 74$, only binomial term $x = 33$ will yield a correct binomial value using this setup. Since 74^2 and 33^2 are congruent mod 4387. Further more calculating the discriminant on polynomial p will yield a square multiple of binomial term x , for example, $DISC(p) = (2 \times x)^2$ if we have a correct solution. See previous sub-chapter.

The main issue with this setup when performing binomial expansion using $BE(x - y)^e$, is that when expanding to even exponents, we need to know which multiple of N to substitute with. Because this will determine whether or not a root solution exists, derived from the binomial terms, that will evaluate the Quadratic to 0. Hence using this oracle setup we can restate the factorization problem as finding a multiple of N to substitute the constant value in $BE(x - y)^e$, where e is even, with. Luckily for us, we can calculate the possible multiples of N modulo m (where m is a product of primes). And this is what I propose as an improvement in the next chapter (Chapter 3 – Improvements).

d. An extended example

Let $N = 4387$

Let binomial $b = x - 534$.

$$\text{Hence } BE(x - 534)^2 = x^2 - 1068x + 534^2$$

We now need to substitute the constant in the resulting polynomial with a multiple of N . For example let multiple $k = 1$.

$$k = 1 \rightarrow p = x^2 - 1068x + 4387$$

$DISC(p)$ is not a square in $\mathbb{Z}$ hence we know $k = 1$ cannot be a valid solution.

However if $k = 65$:

$$k = 65 \rightarrow p = x^2 - 1068x + 4387 \times 65 \text{ and } DISC(p) = 2^2$$

The discriminant is defined as $DISC(p) = (2 \times x)^2$, where x is the binomial term, we can now substitute this into b .

$b(1) = 1 - 534 = -533$ and $p(533) = 0$. Since p evaluates to 0 in $\mathbb{Z}$ we know we have a correct solution and we can take the gcd using our binomial terms:

$$\begin{aligned} \text{GCD}(1 + 534, 4387) &= 107 \\ \text{GCD}(534 - 1, 4387) &= 41 \end{aligned}$$

In the next chapter we will see how to find these solutions in $\mathbb{Z}/p\mathbb{Z}$

Chapter 3 – Improvements

a. Calculating polynomial residues in $\mathbb{Z}/p\mathbb{Z}$

Let us restrict ourselves to binomial expansions to the second degree.

The way we calculate polynomial residues is that we must consider all possible coefficients in $\mathbb{Z}/p\mathbb{Z}$ whose discriminant evaluates to a quadratic residue.

Here we calculate all possible residues for the quadratic polynomial $ax^2 - bx + Nk$ where $N = 4387$ (you can also calculate them for $ax^2 + bx - Nk$ instead, it just changes the side we are calculating in the expression $x^2 = y^2 \pmod N$). We will ignore the cases where the leading coefficient is 0, as this would downgrade the degree of our polynomial. And in addition we will also ignore the case where $k = 0$ for the sake of brevity.

Mod 3

$$k = 1, a = 1 \text{ and } b = -1 \rightarrow x = \{2\}, k = 1, a = 1 \text{ and } b = -2 \rightarrow x = \{1\}, k = 1, a = 2 \text{ and } b = 0 \rightarrow x = \{1, 2\}$$

$$k = 2, a = 1 \text{ and } b = 0 \rightarrow x = \{2, 1\}, k = 2, a = 2 \text{ and } b = -1 \rightarrow x = \{1\}, k = 2, a = 2 \text{ and } b = -2 \rightarrow x = \{2\}$$

Mod 5

$$\begin{aligned} k = 1, a = 1 \text{ and } b = -2 \rightarrow x = \{4, 3\}, k = 1, a = 1 \text{ and } b = -3 \rightarrow x = \{2, 1\}, k = 1, a = 2 \text{ and } b = 0 \rightarrow x = \{3, 2\}, \\ k = 1, a = 2 \text{ and } b = -1 \rightarrow x = \{4\}, k = 1, a = 2 \text{ and } b = -4 \rightarrow x = \{1\}, k = 1, a = 3 \text{ and } b = 0 \rightarrow x = \{1, 4\}, \\ k = 1, a = 3 \text{ and } b = -2 \rightarrow x = \{2\}, k = 1, a = 3 \text{ and } b = -3 \rightarrow x = \{3\}, k = 1, a = 4 \text{ and } b = -1 \rightarrow x = \{3, 1\}, \\ k = 1, a = 4 \text{ and } b = -4 \rightarrow x = \{4, 2\} \end{aligned}$$

$$\begin{aligned} k = 2, a = 1 \text{ and } b = 0 \rightarrow x = \{4, 1\}, k = 2, a = 1 \text{ and } b = -1 \rightarrow x = \{3\}, k = 2, a = 1 \text{ and } b = -4 \rightarrow x = \{2\}, \\ k = 2, a = 2 \text{ and } b = -1 \rightarrow x = \{2, 1\}, k = 2, a = 2 \text{ and } b = -4 \rightarrow x = \{3, 4\}, k = 2, a = 3 \text{ and } b = -2 \rightarrow x = \{3, 1\}, \\ k = 2, a = 3 \text{ and } b = -3 \rightarrow x = \{4, 2\}, k = 2, a = 4 \text{ and } b = 0 \rightarrow x = \{2, 3\}, k = 2, a = 4 \text{ and } b = -2 \rightarrow x = \{4\}, \\ k = 2, a = 4 \text{ and } b = -3 \rightarrow x = \{1\} \end{aligned}$$

$$\begin{aligned} k = 3, a = 1 \text{ and } b = 0 \rightarrow x = \{3, 2\}, k = 3, a = 1 \text{ and } b = -2 \rightarrow x = \{1\}, k = 3, a = 1 \text{ and } b = -3 \rightarrow x = \{4\}, \\ k = 3, a = 2 \text{ and } b = -2 \rightarrow x = \{2, 4\}, k = 3, a = 2 \text{ and } b = -3 \rightarrow x = \{1, 3\}, k = 3, a = 3 \text{ and } b = -1 \rightarrow x = \{3, 4\}, \\ k = 3, a = 3 \text{ and } b = -4 \rightarrow x = \{2, 1\}, k = 3, a = 4 \text{ and } b = 0 \rightarrow x = \{4, 1\}, k = 3, a = 4 \text{ and } b = -1 \rightarrow x = \{2\}, \\ k = 3, a = 4 \text{ and } b = -4 \rightarrow x = \{3\} \end{aligned}$$

$$\begin{aligned} k = 4, a = 1 \text{ and } b = -1 \rightarrow x = \{2, 4\}, k = 4, a = 1 \text{ and } b = -4 \rightarrow x = \{1, 3\}, k = 4, a = 2 \text{ and } b = 0 \rightarrow x = \{4, 1\}, \\ k = 4, a = 2 \text{ and } b = -2 \rightarrow x = \{3\}, k = 4, a = 2 \text{ and } b = -3 \rightarrow x = \{2\}, k = 4, a = 3 \text{ and } b = 0 \rightarrow x = \{2, 3\}, \\ k = 4, a = 3 \text{ and } b = -1 \rightarrow x = \{1\}, k = 4, a = 3 \text{ and } b = -4 \rightarrow x = \{4\}, k = 4, a = 4 \text{ and } b = -2 \rightarrow x = \{1, 2\}, \\ k = 4, a = 4 \text{ and } b = -3 \rightarrow x = \{3, 4\} \end{aligned}$$

The problem is, that we can Chinese remainder these together to modulo 15, but then the solution set of possible residues grows quickly (the product of the size of both sets of solutions). So we need to come up with a smart way to generate solutions with this setup.

b. Singular quadratic roots vs non-singular quadratic roots

In the above residue mapping, we will observe that sometimes a set of coefficients will have a single root solution, and sometimes it will have two root solutions.

For example, looking in Mod 5 we see the following two coefficient, root pairings:

$$\begin{aligned} k = 1, a = 4 \text{ and } b = -1 \rightarrow x = \{3, 1\} \text{ Non-singular roots} \\ k = 1, a = 3 \text{ and } b = -2 \rightarrow x = \{2\} \text{ Singular root} \end{aligned}$$

Singular roots indicate that the corresponding binomial term has prime p (5 here) as divisor and that the discriminant is also divided by prime p .

We have calculated the above residues using polynomials of the shape $ax^2 - bx + Nk$.

If we calculate the derivative in $\mathbb{Z}/5\mathbb{Z}$ we get the residue for linear coefficient b in $\mathbb{Z}/5\mathbb{Z}$ corresponding to a quadratic with the signs flipped $aax^2 + bx - Nk$ (the other side of the expression $x^2 = y^2 \pmod N$ that we are trying to solve for).

Calculating the derivative for coefficients with a singular root in $\mathbb{Z}/p\mathbb{Z}$ gives us a derivative that is 0 in $\mathbb{Z}/p\mathbb{Z}$. Since the derivative calculates the linear coefficient for the other side of the expression $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$, and since the linear coefficient is constructed by multiplying the binomial term by 2, any divisors of the linear coefficient are hence also divisors of the binomial term.

For example using: $k = 1, a = 3$ and $b = -2 \rightarrow x = \{2\}$ we construct polynomial $p = 3 \times 2^2 - 2 \times 2 + 2$ in $\mathbb{Z}/5\mathbb{Z}$. And we see that the derivative $p' = 2 \times 3 \times 2 - 2 = 0 \pmod 5$ (or in other words 0 in $\mathbb{Z}/5\mathbb{Z}$).

This implies that the binomial term y , in binomial $b = y - x$ equals to 0 in $\mathbb{Z}/5\mathbb{Z}$.

Non-singular roots always yield non-zero residues for the binomial terms in $\mathbb{Z}/p\mathbb{Z}$.

c. p-adic lifting of polynomials in $\mathbb{Z}/p\mathbb{Z}$

Let us say we have the following coefficient and root pairing from our residue mapping:

$$k = 4, a = 3 \text{ and } b = -1 \rightarrow x = \{1\} \text{ in } \mathbb{Z}/5\mathbb{Z} \text{ gives us polynomial } p = 3 \times 1^2 - 1 \times 1 + 4387 \times 4 = 0 \pmod 5$$

This gives us all of the following possible solutions in $\mathbb{Z}/25\mathbb{Z}$ where also the derivative yields 0 in $\mathbb{Z}/25\mathbb{Z}$, for the sake of brevity we will focus on these cases only:

$k = 4, a = 3$ and $b = -1 \rightarrow x = \{21\}$, $k = 4, a = 8$ and $b = -6 \rightarrow x = \{16\}$, $k = 4, a = 13$ and $b = -11 \rightarrow x = \{11\}$
 $k = 4, a = 18$ and $b = -16 \rightarrow x = \{6\}$, $k = 4, a = 23$ and $b = -21 \rightarrow x = \{1\}$

$k = 9, a = 3$ and $b = -11 \rightarrow x = \{6\}$, $k = 9, a = 8$ and $b = -16 \rightarrow x = \{1\}$, $k = 9, a = 13$ and $b = -21 \rightarrow x = \{21\}$
 $k = 9, a = 18$ and $b = -1 \rightarrow x = \{16\}$, $k = 9, a = 23$ and $b = -6 \rightarrow x = \{11\}$

$k = 14, a = 3$ and $b = -21 \rightarrow x = \{16\}$, $k = 14, a = 8$ and $b = -1 \rightarrow x = \{11\}$, $k = 14, a = 13$ and $b = -6 \rightarrow x = \{6\}$
 $k = 14, a = 18$ and $b = -11 \rightarrow x = \{1\}$, $k = 14, a = 23$ and $b = -16 \rightarrow x = \{21\}$

$k = 19, a = 3$ and $b = -6 \rightarrow x = \{1\}$, $k = 19, a = 8$ and $b = -11 \rightarrow x = \{21\}$, $k = 19, a = 13$ and $b = -16 \rightarrow x = \{16\}$
 $k = 19, a = 18$ and $b = -21 \rightarrow x = \{11\}$, $k = 19, a = 23$ and $b = -1 \rightarrow x = \{6\}$

$k = 24, a = 3$ and $b = -16 \rightarrow x = \{11\}$, $k = 24, a = 8$ and $b = -21 \rightarrow x = \{6\}$, $k = 24, a = 13$ and $b = -1 \rightarrow x = \{1\}$
 $k = 24, a = 18$ and $b = -6 \rightarrow x = \{21\}$, $k = 24, a = 23$ and $b = -11 \rightarrow x = \{16\}$

Each of these solutions map to the same solution in $\mathbb{Z}/5\mathbb{Z}$. We see a total of 25 possible solutions in $\mathbb{Z}/25\mathbb{Z}$ generated from a single solution in $\mathbb{Z}/5\mathbb{Z}$. I refer to p-adic lifting, but usually it is used in the context of lifting roots of polynomials, but in the context that I use this term, it is also used to lift coefficient residues in $\mathbb{Z}/p\mathbb{Z}$ p-adically. So please keep that in mind. I tend to create these abstractions in my head and the names I give these abstract concepts may not be strictly correct in formal mathematical terms. But to me it would seem, that whether we are lifting root residues or coefficient residues, the underlying mechanism are very similar and connected.

D. Converting singular roots to non-singular roots and visa versa

Because we are searching for a two squares such that $x^2 = y^2 \pmod N$, and since both x and y can also be represented as binomials which can be expanded to the second degree, we can substitute the expression $x^2 = y^2 \pmod N$ with the more elaborate: $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$.

This also means that we can convert singular roots to non-singular roots, simply by transferring a root to the other side of the equation in $\mathbb{Z}/p\mathbb{Z}$. As long as either b_0 or b_1 does not have p as divisor.

For example $1x^2 + 66x - 4387 = 0 \pmod{37}$ has a singular root, while the converse $1x^2 - 0x + 4387 = 0 \pmod{37}$ has two roots.

$$b_1 = 66 \rightarrow 1 \times 4^2 + 66 \times 4 - 4387 = 0 \pmod{37}$$

$$b_0 = 0 \rightarrow 1 \times 4^2 - 0 \times 33 + 4387 = 0 \pmod{37}$$

$$b_0 = 0 \rightarrow 1 \times 33^2 - 0 \times 33 + 4387 = 0 \pmod{37}$$

But if we flip the sign on the linear coefficient b_1 , then the singular root becomes the other non-singular root for b_0 (we must also swap a and k , but both are 1 here):

$$b_1 = -66 \rightarrow 1 \times 33^2 - 66 \times 33 - 4387 = 0 \pmod{37}$$

Chapter 4 – An algorithm outline

Traditional Quadratic Sieve algorithms are mainly bottle-necked by the fact that beyond 110 digits the factor base required for successful factorization becomes too large. This is mainly due to the relation between factor base size and the on average number of B -smooths required to succeed at the linear algebra step. Self Initializing Quadratic Sieve has some improvements, the biggest one being the speed it up gets by guaranteeing the factorization of $\pm$ half of the B -smooth candidates being generated. And it also slightly improves our chances of succeeding at linear algebra earlier, as there will be more overlap between factors. SIQS is somewhat the cutting edge of Quadratic Sieve Variants. The next logical improvement would be to achieve more overlap between factors to make the chance of succeeding at the linear algebra step less dependent to the factor base size. For example, if we have a factor base size of 10000 prime numbers and are able to succeed with only 100 B -smooths. A new variant like this, would likely also out perform the number field sieve algorithm and give us a real attempt at RSA-1024.

There is a way using binomial expansions that something like this can be achieved.

Let us attempt to generate B -smooth candidates for $N = 431272907527067$

Step 1 - Binomial expansion to degree d

We iterate y for the binomial $x - y$ and expand to degree d .

Setting $y = 1$ and expanding to degree $d = 2$ we get:

$$BE(x - 1)^2 = x^2 + 2x + 1$$

Step 2 - sieving

We sieve using a two dimensional interval, one dimension will sieve the roots, which will be multiples of the binomial value and the other dimensions will be multiples of N that we substitute the constant with.

For example if $x = 1 \times 168$ and the multiplier to N , $k = 229235148952305$

Then polynomial $p = x^2 + 2x + 431272907527067 \times 229235148952305$, where we substitute the constant in the binomial expansion from step 1 with Nk .

And evaluating p with x , $p(x) = 168^2 + 2 \times 168 + 431272907527067 \times 229235148952305 = 98862909196060863953479567995$ which factorizes as:

$$p(x) = 3 \times 5 \times 47 \times 53 \times 103 \times 157 \times 223 \times 283 \times 311 \times 313 \times 367 \times 379 \times 419 \times 457$$

If factor base B is all primes up to 1000, then this polynomial value is B -smooth. Since all its prime factors are less than 1000.

In addition, since we are sieving with polynomial roots that have a non-zero derivative, this also means that the part without the constant must also factorize over the factor base, as it is not guaranteed to be square, and we must end up with a square on this side as well after the linear algebra step. We will call this side the left side of the square relation and the polynomial value the right side.

$$\text{Factorization of the left side: } 168^2 + 2 \times 168 = 28560 = 2 \times 2 \times 2 \times 2 \times 3 \times 5 \times 7 \times 17$$

We sieve multiples of the binomial value, because as long as the binomial value factorizes, this guarantees that what remains on the left side is relatively small if we sieve for examples multiples of the binomial value 1 to 1000. This helps us ensure that we only need to really worry about the factorization on the right side (the polynomial value). It is not a requirement to do it this way, but from a practical point of view it simplifies things.

Step 3 - finding variations of a B-smooth

In the previous step we found a B -smooth. If we can find that same B -smooth again, but with a few different factors, then all the factors that are the same cancel each-other out during the linear algebra step. This can then potentially help us succeed much sooner.

The way this is now possible is due to the following. Lets say we are sieving with binomial expansions of binomial $b = x + y$ to degree $d = 2$ and substituting the constant with a multiple of N defined by multiplier k , giving us the following quadratic:

$$BE(x + y)^2 = ax^2 + b_1x - Nk.$$

We check values for x and k , x modifies the value of $ax^2 + b_1x$ and k modifies the value of Nk . The trick is to find a combination of a root x and constant multiplier k such that the result is a B -smooth of near identical factorization.

For example if $x = 1 \times 238$ (where 1 is the binomial term) and $k = 458470297904610$ (note that this is double the k value that was used to find the previous B -smooth from step 2)

$$\text{Hence } p = x^2 + 2x + 431272907527067 \times 458470297904610$$

$$\text{And } p(x) = 238^2 + 2 \times 238 + 431272907527067 \times 458470297904610 = 197725818392121727906959135990 \text{ which factorizes as:}$$

$$p(x) = 2 \times 3 \times 5 \times 47 \times 53 \times 103 \times 157 \times 223 \times 283 \times 311 \times 313 \times 367 \times 379 \times 419 \times 457$$

Which is the same B -smooth as before with 2 added as a factor. Both of these B -smooths can be added to the linear algebra step without problems. The only problematic case is when the root x is multiplied by some value v and k by the square of v , in essence adding v^2 to the polynomial value. Which counts as a multiple of the original smooth. And these cases count as duplicates. But in this case, the roots are not multiples of each-other and we are only adding a single non-square factor to the polynomial value.

Further more, when trying to find these similar B -smooths with near identical factorization, which you would do by looking at multiples of a k value from the original B -smooth, we should first divide out common factors in k and the polynomial value. And look for multiples of this instead.

For example in our original example if $x = 1 \times 168$ and $k = 229235148952305$

$$\text{Our polynomial } p = x^2 + 2x + 431272907527067 \times 229235148952305$$

$$\text{And } p(x) = 168^2 + 2 \times 168 + 431272907527067 \times 229235148952305 = 98862909196060863953479567995 \text{ which factorizes as:}$$

$$p(x) = 3 \times 5 \times 47 \times 53 \times 103 \times 157 \times 223 \times 283 \times 311 \times 313 \times 367 \times 379 \times 419 \times 457$$

Here both k and $p(x)$ have 5 as common factor. So rather than looking for small multiples of $p(x)$ we should look for multiples of $p(x) / 5$ and $k / 5$ instead.

When we do this, we find two more B -smooths of near identical factorization:

$$\text{If } x = 406 \text{ and } k = 1329563863923369 \text{ (note that this is } 29 \times (229235148952305 / 5) \text{)}$$

$$\text{Hence } p = x^2 + 2x + 431272907527067 \times 1329563863923369$$

$$\text{And } p(x) = 406^2 + 2 \times 406 + 431272907527067 \times 1329563863923369 = 573404873337153010930181494371 \text{ which factorizes as:}$$

$$p(x) = 3 \times 29 \times 47 \times 53 \times 103 \times 157 \times 223 \times 283 \times 311 \times 313 \times 367 \times 379 \times 419 \times 457$$

This is our original polynomial value, divided by 5 and multiplied by 29. Hence another slight variation of our original B -smooth.

And a final example:

$$\text{If } x = 544 \text{ and } k = 2384045549103972 \text{ (note that this is } 52 \times (229235148952305 / 5) \text{)}$$

$$\text{Hence } p = x^2 + 2x + 431272907527067 \times 2384045549103972$$

$$\text{And } p(x) = 544^2 + 2 \times 544 + 431272907527067 \times 2384045549103972 = 1028174255639032985116187507148 \text{ which factorizes as:}$$

$$p(x) = 2 \times 2 \times 3 \times 13 \times 47 \times 53 \times 103 \times 157 \times 223 \times 283 \times 311 \times 313 \times 367 \times 379 \times 419 \times 457$$

This is our original polynomial value, divided by 5 multiplied by 52. Hence, yielding another slight variation of the original B -smooth.

From a single B -smooth we were quite trivially able to find 3 more B -smooths with very similar factorization.

The existence of those can easily be checked for using residue math as we simply multiply k by some value whose factors are within the factor base such that this value gets added to the polynomial value while keeping the original B -smooth intact. This behavior is aided by the fact that the left side of the square relation is not forced to be strictly square, hence behaving as a linear offset and k , the multiplier to the constant behaving as another linear offset.

This demonstrates how it is possible to get an almost full match with the factorization of our B -smooths, rather than the partial match of factors SIQS achieves. Further more, this same process can be repeated over and over again by expanding to higher degrees, which is also not something seen in other QS variants.

Note: I am still working on the PoC, and until I can actually demonstrate it succeeding at the linear algebra step with much fewer smooths, still remains simply an untested conjecture. But the math on this looks good so far. Will finalize this paper once the code is done, and this goes from untested conjecture to tested conjecture.. and perhaps even some new theorems can come out of this one day.